

An Adjustment Strategy On Multi-Session EEG Data for Online Left/Right Hand Imagery Classification

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Abstract—In this research, electroencephalography (EEG) is used as an interface to communicate between patients and doctors. The signals from two electrodes (C3 and C4) are captured and used to classify Left/Right hand imagery representing YES/NO answers of the patients. In online applications, the training model mostly cannot be applied to the testing sessions due to a variation of the signals. Although some prior works employed a normalization technique, the parameters were still derived from all sessions, not just the training sessions, resulting in low prediction accuracy in real-world online systems. In this paper, we propose an adjustment strategy that can be applied online to all features by subtracting “estimated mean” and dividing “estimated interquartile range” (IQR) or “estimated standard deviation” (SD), which are obtained by using exponentially weighted moving average (EWMA). In our system, the features are extracted by applying the wavelet transformation, and Neural Network is chosen as our classifier. The experiment was conducted on the BCI IV data set and compared to four existing techniques: (i) non-normalized wavelet, (ii) Z-transform, (iii) CSP, and (iv) CSP with Morlet wavelet, in terms of accuracy. The results showed that our proposed method significantly outperformed the first three works and it is comparable to the last one, but ours employed the less number of electrodes.

Keywords—component; Brain Computer Interface (BCI); Electroencephalogram (EEG); Motor imagery (MI); Neural Network; Wavelet Transform; Exponential Weight Moving Average (EWMA)

I. INTRODUCTION

Locked-in Syndrome (LIS) [1] is a condition, in which a patient is aware and conscious, but cannot move or communicate verbally due to complete paralysis of nearly all voluntary muscles in the body except for the eyes. In addition, Total locked-in Syndrome is a more troubling condition where the eyes are also paralyzed. Since LIS is caused by damage of specific portions of the lower brain and there is no damage of the upper brain, the patients can still communicate using a Brain Computer Interface (BCI) system. BCI is a direct communication pathway between the brain and an external device. In this study, electroencephalography (EEG) will be used as an interface since it is non-invasive, easy to use, and has low setup cost and fine temporal resolution.

In multi-sessions classification, EEG signals have inherent non-stationary characteristics. Therefore, BCI features vary

from training sessions to testing sessions [2]. Most research eliminate the possibility of high variance features dominating the classification training by rescaling to have zero mean and unit standard deviation (SD) [3]. There are many successful classification results using this normalization method *after* feature extraction as shown in [4-7]. Unfortunately, they require the global mean and global SD of all extracted features of the testing data, which are unknown in the online system. Some prior attempts apply the normalization directly to the EEG signal in the time domain *before* feature extraction [8-10] which is the most common calibration method used in the online prediction system and considered as our baseline. There are also other normalization techniques, such as range transformation (scaling data based on the min/max range) and proportion transformation (scaling data based on the total sum). However, they are not included in this paper, since they are sensitive to outliers and not suitable for the brain signals.

For the choices of electrodes, the previous knowledge tells us that the C3, Cz and C4 electrodes, which are over the sensorimotor area, can capture important characteristics of motor imagery [11-13]. Ge, et al. [11] applied short time Fourier transform (STFT) to decompose a single channel EEG signal into multiple signals (mu and beta bands) and used them to reconstruct the signal. This work presented the best result at C4 and showed that the result of C3 was better than Cz. Therefore, the EEG locations are minimized to be used C3 and C4 in our work.

For the feature extraction, there has been a large number of published works related to the techniques for Motor Imagery (MI) classification of EEG. The wavelet transformation is one of the most common feature extractions used in EEG classification [14-19]. Also, many recent works in 2015 [2, 4, 8, 20-23] chose Common Spatial Pattern (CSP) as their feature extraction. Later Bashashati, et al. [19] proposed to cooperate CSP with Morlet wavelet transform and showed promising results. However, CSP only performed well with multiple EEG channels. In this work, the wavelet method is our choice of feature extraction. Since CSP is one of the most popular method used in EEG classification, we will use “Filter Bank CSP” (FBCSP) and “Filter Bank CSP with Morlet wavelet” as our baselines.

In this paper, we propose an adjustment strategy to normalize the EEG features during online Left/Right hand

imagery prediction. After processing wavelet transformation, the extracted features are subtracted by the “estimated mean value” and divided by the “estimated Inter-Quartile Range (IQR)” or “estimated SD” of the current session. Finally, the normalized features are fed to Neural Network classifier to predict either YES or No. The experiment was conducted on a standard benchmark, the BCI Competition IV data set, and compared the result to four baselines including (i) non-normalized wavelet, (ii) Z-transform normalization directly to the EEG signal, (iii) CSP, and (iv) CSP with Morlet wavelet.

The rest of the paper is organized as follows: Section II describes the data set used in this study. Section III presents details of the proposed approach. Section IV describes the experimental setup. Section V provides the experimental results and discussion. Finally, Section VI gives a conclusion of the study.

II. BCI COMPETITION DATA SET

The proposed method is evaluated on the BCI Competition IV on the data set 2B provided by Graz University [24]. It contains EEG signals from nine subjects performing two different imagery tasks Left/Right hands. There are three sessions for training data without smiley feedback and two sessions of evaluation data with smiley feedback; the subjects were required to move the smiley towards the left or right side by imagining left or right hand movements. Each session includes 90 training trails and 160 evaluation trails, except subject 2 with 140 evaluation trails. The signals were sampled at 250 Hz and applied the band-pass filter between 0.5 Hz and 100 Hz. A 50 Hz notch filter was also applied to suppress the power line noise. All volunteers were sitting in an armchair, watching at a monitor. For the training session, a cross was shown on the black screen and a short warning tone was given at the beginning of each trial. Then, an arrow pointing to either left or right side was presented. Finally, the subjects were prompted to perform the corresponding motor imagery task over a period of four seconds. For the evaluation session, the feedback (a gray smiley) was centered on the screen and a short warning beep was given at the beginning of each trial. Then, cue was presented over a period of four seconds. Finally, the subjects were required to move the smiley towards the left or right side by imagining left or right hand movements respectively.

III. PROPOSED METHOD

Our objective is to introduce an adjustment method that minimizes the variation among multiple sessions of EEG data and can be employed in the online prediction system. In our system, there are four steps as shown in Fig. 1.

A. Step1: Data Segmentation

For each trial, the signal is segmented by removing the beginning and ending parts during the transition state. So, the signal is selected at the 3.5th until the 6.5th seconds (three seconds of the total window size) to be processed by the wavelet transformation.

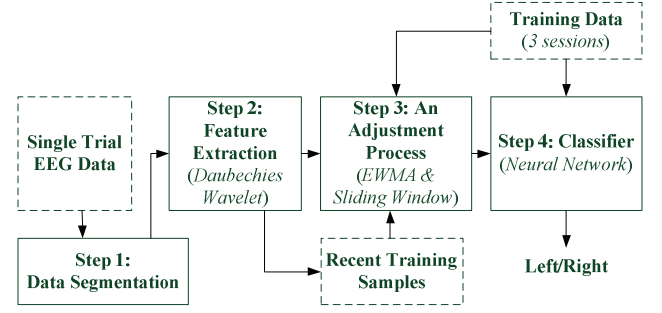


Figure 1. A process flow of the proposed algorithm

B. Step2: Feature Extraction

For each electrode, a signal is extracted into three frequencies bands including alpha (8-13.9Hz), beta (14-30Hz) and gamma (more than 30Hz, but limited to 45Hz). Since there are two electrodes (C3 and C4), so there are total six wavelet features that will be used in the classification process.

C. Step3: An Adjustment Process

There is usually a discrepancy among different sessions resulting in low prediction performance. In this work, an adjustment strategy is proposed to eliminate this issue by using a normalization technique as shown in the equation below. At time t of i -th feature, the normalized value ($normalize(x(i,t))$) can be computed by two parameters: $mean(i,t)$ and $range(i,t)$.

$$normalize(x(i,t)) = (x(i,t) - mean(i,t)) / range(i,t) \quad (1)$$

For the mean parameter, it can be computed by using the exponential weighted moving average technique (EWMA) [25] as shown in the equation below. At time t of i -th feature, the smoothed mean ($mean(i,t)$) is a weighted sum of the previous mean ($mean(i,t-1)$) and the current value ($x(i,t)$), where η is an update coefficient factor, it is set to 0.1 in this work.

$$mean(t) = (1 - \eta) \cdot mean(i,t-1) + \eta \cdot x(i,t) \quad (2)$$

For the range parameter, there are two choices, i.e., standard deviation (SD) and inter-quartile range (IQR). Their estimated value can be computed by applying the EWMA technique and sliding window as shown in the following equation. At time t of i -th feature, the smoothed range ($range(i,t)$) is a weighted sum of the previous range ($range(i,t-1)$) and the current range of the window of size k at time between $(t-k)$ and t ($range(i, t-k \rightarrow t)$). In the experiments, the window size is fixed to 20.

$$range(t) = (1 - \eta) \cdot range(i,t-1) + \eta \cdot range(i,t-k \rightarrow t) \quad (3)$$

D. Step4: Classification

In the last step, a classifier, Neural Network (NN), is induced using the normalized wavelet data. There are two kinds of models: “subject specific model” and “universal model.” The first model is built separately for each subject, while the latter one is built only once and used for all subjects.

IV. EXPERIMENTAL SETUP

In the experiments, there are eight techniques comprising of four variations of the normalization techniques (including our proposed method) and four baselines.

For the normalization approach, it requires mean and range to adjust the data. Different methods propose various solutions to obtain these two parameters. There are four normalization-based approaches. The first method is called “20-trial-adjust,” which estimates the parameters using the first 20 trials. The second method is called “30-trial-adjust,” which is similar to the first method, but depends on the first 30 trials. The third method is our proposed solution (Section III), which is abbreviated as “auto adjust”. Furthermore, the last method is called “limit adjust,” which is similar to the previous solution (“auto adjust”), but the number of adjustment in the same direction is limited to prevent the system from giving a single output (either Left or Right for the whole testing data). Note that the first and second methods assume that there is no change in the adjustment parameters (mean and range) within the same session, so their estimated parameters are fixed and used for the whole testing data. However, this assumption may not be true, so the third and fourth solutions allow that the adjustment parameters can be updated within the same session.

There are four baseline techniques that are commonly used in prior works. First, “Filter Bank CSP” is selected as our baseline, which is a combination of 4th order of Butterworth and Spatial filters. Second, “Filter Bank CSP with Morlet Wavelet” is similar to the traditional CSP cooperating with Morlet wavelet transform [19]. Third, it applies “Z-transform” to each trail of the sampled EEG signal. Finally, there is no normalization among wavelet signals from different sessions; it is called “non-normalize.”

All of the eight techniques above were experimented on the BCI Competition IV (Section 2). The data set contains five sessions of nine subjects. For each subject, the data is separated into training and testing data of three and two sessions, consecutively. The t-test was conducted to compare the significant difference between each pairs of models.

V. RESULTS AND DISCUSSIONS

A. A comparison of user-specific models using the proposed methods

In this section, four normalization methods (including ours) are compared as shown in Table I. We also compare between the results of adjusting “mean” and both “mean and range,” where there are two options of ranges: “IQR” and “SD”.

The results show that the winner is our proposed method (“auto adjust”) with an adjustment of mean and SD (8th

column). The overall accuracy is 76.45±13.52%. It outperforms our strategy with only mean adjustment (1st column) for 1%, so the range parameter is required in the normalization. For the choice of range, SD is always better than IQR by comparing it to Column 6th.

Also, our method (“auto adjust”) surpasses “20-trial adjust” and “30-trial adjust,” (2nd and 3rd columns). Not only our methods, all other auto adjusted strategies are better than both fixed adjustment methods. This can be concluded that the auto adjusted normalization is suitable for EEG signals than the fixed adjusted normalization. Moreover, the result of “30-trial adjust” is always better than that of “20-trial adjust”.

TABLE I. ACCURACY OF SUBJECT-SPECIFIC MODELS ON DIFFERENT NORMALIZATION METHODS. WINNERS ARE BOLD FACED.

Subject	Auto adjust	20-trial-adjust	30-trial-adjust	Limit adjust	Auto adjust	Limit adjust	Auto adjust
	Mean	Mean and IQR				Mean and SD	
1	72.81	63.75	64.06	65.93	69.37	72.18	70.62
2	55.35	57.14	57.14	55.00	56.07	62.14	62.50
3	52.81	53.75	56.25	54.06	54.37	55.31	55.00
4	96.56	83.13	85.93	95.00	96.56	94.68	95.00
5	78.13	86.87	88.75	90.31	88.75	88.43	88.43
6	79.06	77.81	79.68	78.75	77.50	77.50	76.87
7	69.37	70.31	70.62	68.13	70.62	63.75	71.56
8	91.56	87.50	87.81	87.81	88.75	88.43	89.06
9	79.68	82.18	82.50	81.87	81.87	80.31	79.06
Average	75.03	73.60	74.75	75.20	75.98	75.85	76.45
SD	14.58	12.86	13.03	15.11	14.67	13.52	13.06

B. A comparison of user-specific models between the proposed method and baselines

In this section, our method is compared to four baselines. Note that the results of “Filter Bank CSP with Morlet wavelet” is obtained from the source paper [19] which using Multi-Layer Perceptron Classifier. Table II shows that the proposed method outperforms all baseline techniques. It is significantly better than “Z-transform” ($p=0.0015$), “Filter Bank CSP” ($p=0.0204$), and “non-normalize” ($p=0.0034$). Also, it is comparable to “Filter Bank CSP with Morlet wavelet” ($p=0.4704$), but our algorithm is much simpler and requires the less number of electrodes, only 2 probes (C3 and C4).

TABLE II. ACCURACY OF SUBJECT-SPECIFIC MODELS COMPARING BETWEEN PROPOSED AND BASELINE METHODS. WINNERS ARE BOLD FACED.

Subject	Proposed Method (Adjust Mean and SD)	Baseline Method			
		Filter Bank CSP	FBCSP with Wavelet	Z-transform	Non-normalize
1	70.62	71.25	69.30	63.43	67.50
2	62.50	50.00	58.37	53.92	50.00
3	55.00	47.81	51.74	51.87	50.00
4	95.00	92.50	94.46	85.00	77.81
5	88.43	64.37	85.71	90.00	65.62
6	76.87	50.00	83.27	70.31	50.00
7	71.56	50.00	71.12	67.18	50.00
8	89.06	89.06	80.00	89.37	90.93
9	79.06	77.81	83.67	69.37	50.00
Average	76.45	65.86	75.29	71.16	61.31
SD	13.06	17.68	13.81	14.25	15.19

C. A comparison of user-specific and universal models

In this section, we compare our method between a subject-specific model (a separated model per subject) and the universal model (a single model for all subjects). This comparison is helpful since it allows an opportunity to apply a global model to unknown subjects. Based on the literature survey, the comparison is commonly found in the EEG emotion prediction, not in the Motor Imagery classification. In [26], an accuracy of the subject-specific model is usually better than that of the universal model by 12%.

In our experiment, Table III shows that an accuracy of a subject-specific model is better than that of the universal model for 1%. Using t-test, they are comparable ($p=0.4778$). This result shows that by normalizing the wavelet power feature using our proposed method works well and good enough to build a common model for all subjects.

TABLE III. ACCURACY OF PROPOSED METHOD COMPARING BETWEEN SUBJECT SPECIFIC MODEL AND UNIVERSAL MODEL. WINNERS ARE BOLDFACED.

Subjects	Subject Specific Model	Universal Model
1	70.62	69.30
2	62.50	58.37
3	55.00	51.74
4	95.00	94.46
5	88.43	85.71
6	76.87	83.27
7	71.56	71.12
8	89.06	80.00
9	79.06	83.67
Average	76.45	75.29
SD	13.06	13.81

D. Further Discussions

In this section, we aim to illustrate the reason behind the success of our proposed algorithm, which is based on the wavelet transformation, using the signals of Subject 4 as shown in Fig. 2 and Fig. 3.

Fig. 3 shows the original wavelet power on the C3's beta band (without any adjustments) of two sessions, where Sessions 1 and 2 are represented by circles and triangles, respectively. By observing signals in the first session (circles),

there is a discrepancy of power means within the same session that can be identified into three groups: 1st-60th, 61th-120th, and 121st-160th trials. In the online system, data in the past session (assuming Session 2) are collected to induce a model to predict data in the current session (assuming Session 1). During 121st-160th trials in Fig. 3, there is a difference of means between two sessions, so the model may fail to predict data from other sessions.

Fig. 2 is a set of subplots between two features, where x-axis represents the C3's beta band and y-axis represents the C4's alpha band. There are three subplots showing different normalization techniques. The prediction model aims to classify data into two groups of left-hand imaginary (red) and right-hand imaginary (blue). It is obvious that data cannot be separated in both spaces of the original data (Fig. 2(a)) and the Z-transform on the time-domain data (Fig. 2(b)). By applying the proposed adjustment (Fig. 2(c)), data can be successfully separated into two groups. Moreover, data in the different sessions (represented by circles and triangles) are located in the same region; thus, the model from one session can predict data from different sessions.

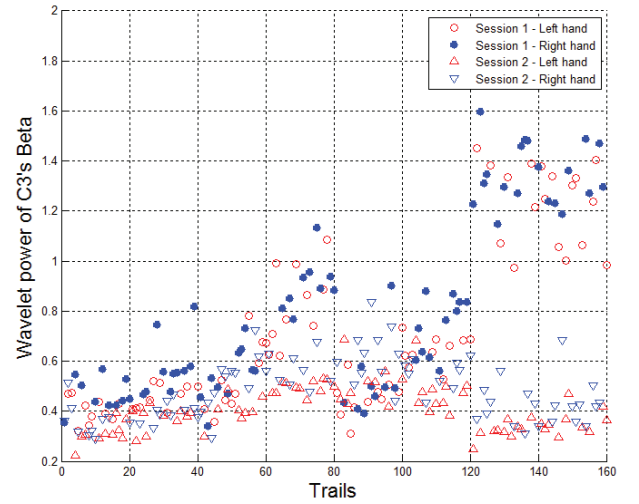


Figure 3. Feature A plot of wavelet power on the C3's beta band among different trails on two sessions of Subject 4

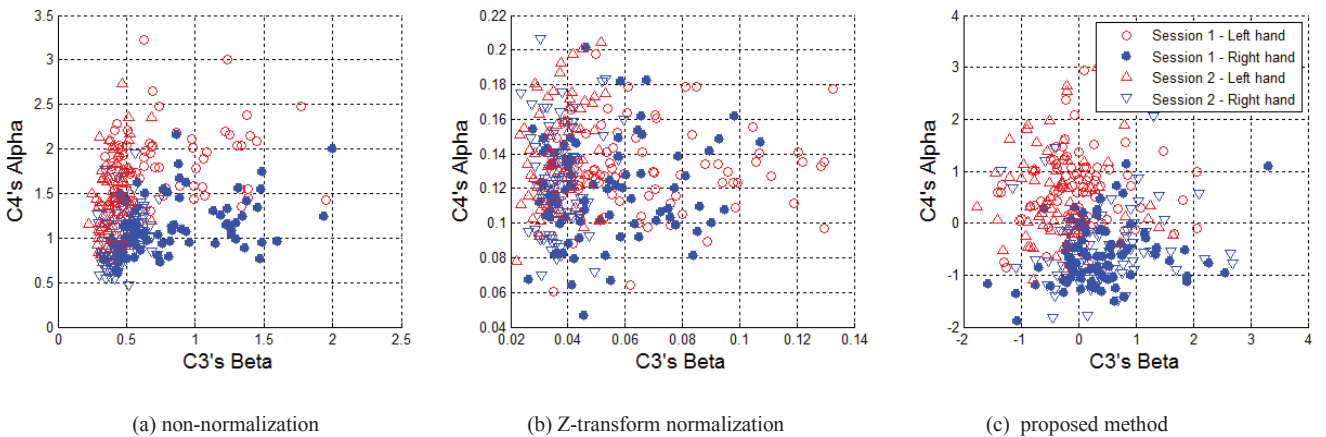


Figure 2. A plot of wavelet power on the C3's beta and C4's alpha bands on two sessions of Subject 4 among three different normalization techniques

VI. CONCLUSION

In this paper, we aim to develop an online classification system of Left/Right hand imagery representing YES/NO answer by using EEG signals gathered from only a pair of channels. The goal of our work is to suggest an adjustment strategy which can be applied online to resolve a discrepancy among multiple sessions of EEG signals. The proposed method is based on a normalization technique that requires two parameters: mean and range (SD or IQR). In the experiment, the BCI Competition IV is chosen. The proposed method is compared to four baselines: (i) non-normalized wavelet, (ii) Z-transform, (iii) CSP, and (iv) CSP with Morlet wavelet, in terms of accuracy. The results show that the proposed method yields better accuracy than all baselines (Z-transform normalization, Filter Bank CSP). It is comparable to the result of Filter Bank CSP cooperating with Wavelet Transform, but it employs the less number of electrodes.

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